

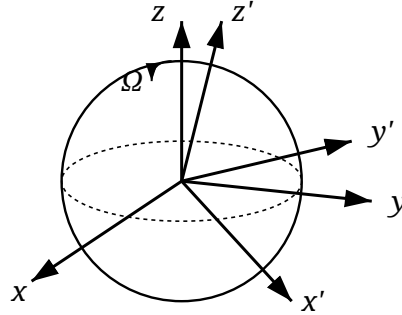


Figure 15.2 A rotating coordinate system (x', y', z') fixed in the Earth and an inertial system (x, y, z) at rest with respect to the fixed stars.

equatorial plane. Again, the coordinate axes (x, y, z) are fixed in space and the coordinate axes (x', y', z') are rigidly attached to the Earth and rotating with it.

A person standing still on the surface of the Earth is at rest with respect to the rotating coordinate system. The position of this person is given by the vector $\mathbf{r}$ from the origin to the point P where the person is standing. The components of $\mathbf{r}$ can be specified in either the inertial or the rotating coordinate system. Because P is at rest in the rotating (Earth or primed) system,

$$\left[\frac{d\mathbf{r}}{dt} \right]_{\text{rot}} = 0.$$

On the other hand, according to an observer in the inertial system, the point P is moving with angular velocity $\boldsymbol{\Omega}$. The linear velocity of the point in the inertial system is

$$\left[\frac{d\mathbf{r}}{dt} \right]_{\text{inertial}} = \boldsymbol{\Omega} \times \mathbf{r}.$$

Next assume the person has a velocity $\mathbf{v}_r$ with respect to the Earth, so that

$$\left[\frac{d\mathbf{r}}{dt} \right]_{\text{rot}} = \mathbf{v}_r.$$

Then, from the point of view of the inertial frame, the person's velocity is

$$\left[\frac{d\mathbf{r}}{dt} \right]_{\text{inertial}} = \mathbf{v}_r + \boldsymbol{\Omega} \times \mathbf{r}.$$

This last equation can be written

$$\left[\frac{d\mathbf{r}}{dt} \right]_{\text{inertial}} = \left[\frac{d\mathbf{r}}{dt} \right]_{\text{rot}} + \boldsymbol{\Omega} \times \mathbf{r}.$$

It is easy to generalize this relationship to the time derivative of any vector $\mathbf{U}$. The relation between the rate of change of $\mathbf{U}$ in the inertial system and the rate of change of $\mathbf{U}$ in the rotating system is

$$\left[\frac{d\mathbf{U}}{dt} \right]_{\text{inertial}} = \left[\frac{d\mathbf{U}}{dt} \right]_{\text{rot}} + \boldsymbol{\Omega} \times \mathbf{U}. \quad (15.3)$$

This can be expressed as an *operator equation*

$$\left[\frac{d}{dt} \right]_{\text{inertial}} = \left[\frac{d}{dt} \right]_{\text{rot}} + \boldsymbol{\Omega} \times, \quad (15.4)$$

where it is understood that the various terms are all operating on the same vector.

Exercise 15.1

Somewhere out in space there is an inertial reference frame and (strange to say) in it there is a merry-go-round that is rotating at angular velocity Ω . A child is sitting on a wooden horse a distance R from the axis of the merry-go-round. (a) What is the velocity of the child in the rotating frame? (b) What is the velocity of the child in the inertial frame? (c) What is the apparent acceleration of the child in the rotating frame? (d) Determine the acceleration of the child in the inertial frame by elementary methods, that is, by using the definition of centripetal acceleration. (e) Determine the acceleration of the child in the inertial frame using [Equation \(15.3\)](#). Answers (a) zero, (b) $\Omega R \hat{\phi}$, (c) zero, (d) $-\Omega^2 R \hat{\rho}$.

Exercise 15.2

Prove that the rate of change of the angular velocity of the rotating system is the same in both the inertial and the rotating reference frame.

15.3 Fictitious Forces

Let us continue ignoring the motion of the Earth around the Sun and imagine the Earth to be a rotating sphere isolated in space. It is rotating at a nearly constant angular velocity of 2π rad/day. The relationship between the time rate of change of the position vector ($\mathbf{r}$) of a particle in the inertial reference frame and its time rate of change in the rotating reference frame is, by [Equation \(15.3\)](#),

$$\left[\frac{d\mathbf{r}}{dt} \right]_{\text{inertial}} = \left[\frac{d\mathbf{r}}{dt} \right]_{\text{rot}} + \boldsymbol{\Omega} \times \mathbf{r}.$$

The velocity of the particle with respect to inertial space is $[d\mathbf{r}/dt]_{\text{inertial}}$. Call it $\mathbf{v}_i$. Also, $[d\mathbf{r}/dt]_{\text{rot}}$ is the velocity of the particle with respect to the rotating frame. Call it $\mathbf{v}_r$. Consequently,

$$\mathbf{v}_i = \mathbf{v}_r + \boldsymbol{\Omega} \times \mathbf{r}.$$

This equation indicates that if a person on the Earth perceives a particle to be at rest, an observer in inertial space will perceive the particle to have a velocity $\mathbf{v}_i = \boldsymbol{\Omega} \times \mathbf{r}$.

Acceleration is defined as the time rate of change of velocity. Hence,

$$\mathbf{a}_{\text{inertial}} = \left[\frac{d\mathbf{v}_i}{dt} \right]_i.$$

This is the acceleration in the inertial reference frame and is, of course, the acceleration that is expressed in Newton's second law. However, we live in a rotating reference frame, so we want to express the second law in terms of quantities we can measure, that is, $\mathbf{a}_r$ and $\mathbf{v}_r$.

Applying operator [equation \(15.4\)](#) yields

$$\begin{aligned} \mathbf{a}_i &= \left[\frac{d}{dt}(\mathbf{v}_r + \boldsymbol{\Omega} \times \mathbf{r}) \right]_{\text{inertial}} \\ &= \left[\frac{d}{dt}(\mathbf{v}_r + \boldsymbol{\Omega} \times \mathbf{r}) \right]_{\text{rot}} + \boldsymbol{\Omega} \times (\mathbf{v}_r + \boldsymbol{\Omega} \times \mathbf{r}), \\ &= \left[\frac{d}{dt}\mathbf{v}_r \right]_{\text{rot}} + \left[\frac{d}{dt}(\boldsymbol{\Omega} \times \mathbf{r}) \right]_{\text{rot}} + \boldsymbol{\Omega} \times \mathbf{v}_r + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}), \\ &= \mathbf{a}_r + \frac{d\boldsymbol{\Omega}}{dt} \times \mathbf{r} + \left[\left(\boldsymbol{\Omega} \times \frac{d\mathbf{r}}{dt} \right) \right]_{\text{rot}} + \boldsymbol{\Omega} \times \mathbf{v}_r + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}). \end{aligned}$$

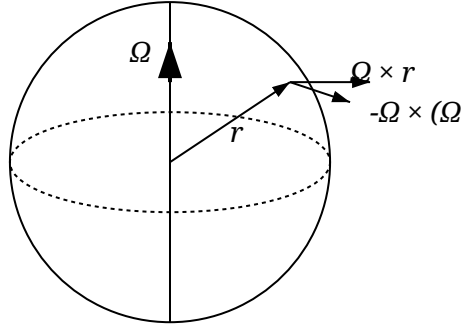


Figure 15.3 The centrifugal acceleration, $-\Omega \times (\Omega \times \mathbf{r})$, points away from the axis of rotation.

If the Earth's rotation rate is assumed constant, $\frac{d\Omega}{dt} = 0$. Using the notation $[d\mathbf{r}/dt]_{\text{rot}} = \mathbf{v}_r$, the inertial acceleration is given by

$$\mathbf{a}_i = \mathbf{a}_r + 2\Omega \times \mathbf{v}_r + \Omega \times (\Omega \times \mathbf{r}).$$

Newton's second law is valid in the inertial frame, so $\mathbf{a}_i = \mathbf{F}/m$, where $\mathbf{F}$ is the net external ("physical") force acting on the particle. Replacing $\mathbf{a}_i$ and rearranging leads to

$$\mathbf{a}_r = \mathbf{F}/m - 2\Omega \times \mathbf{v}_r - \Omega \times (\Omega \times \mathbf{r}). \quad (15.5)$$

This equation shows that the acceleration as *measured in the rotating reference frame* will depend not only on the force acting on the particle, but also on two other terms. These additional terms are a consequence of considering the motion in a noninertial reference frame.

One often writes [Equation \(15.5\)](#) in the suggestive form

$$\mathbf{F} - 2m\Omega \times \mathbf{v}_r - m[\Omega \times (\Omega \times \mathbf{r})] = m\mathbf{a}_r. \quad (15.6)$$

This equation *looks like* $F = ma$. The additional terms on the left-hand side of the equation have the units of force and they are referred to as "fictitious forces." Keep in mind, however, that they really are just mass times acceleration terms that have been brought over from the other side of the equation. A "real" force is an interaction between bodies. A "fictitious" force is a consequence of the acceleration of the coordinate system.

The term

$$-2m\Omega \times \mathbf{v}_r,$$

is called the "Coriolis force" and the term

$$-m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}),$$

is called the "centrifugal force." The reason for the name is that this "force" is directed away from axis of rotation, as illustrated in [Figure 15.3](#)

Worked Example 15.1 An old-fashioned record player is rotating with an angular velocity $\boldsymbol{\Omega}_0 = \Omega_0 \hat{\mathbf{k}}$. A bug is crawling with constant speed v_b relative to the record along a groove in the record, a distance r from the axis of rotation. Assume the bug is moving in a circular path, so there is no radial component to its velocity. (a) Determine the velocity of the bug relative to the room (that is the "inertial space"). (b) Determine the acceleration of the bug by elementary means, using the definition of centripetal acceleration. (c) Determine the acceleration of the bug using [Equation \(15.4\)](#) and compare your answer to part (b). (Comment: This example will help you learn the mechanics of using [Equation \(15.4\)](#).) In particular, you will appreciate that the same vector must be used on both sides of that equation. That is, if you are calculating

$\left[\frac{d\mathbf{v}_i}{dt} \right]_{\text{inertial}}$ you get the wrong answer if you put $\left[\frac{d\mathbf{v}_{\text{rot}}}{dt} \right]_{\text{rot}}$ on the other side. This is fairly obvious, but it is an easy error to make.)

Solution Let $\hat{\boldsymbol{\rho}}, \hat{\boldsymbol{\phi}}, \hat{\mathbf{k}}$ be cylindrical unit vectors in the rotating frame. (a) The velocity of the bug in the rotating coordinate frame is

$$\left[\frac{d\mathbf{r}}{dt} \right]_{\text{rot}} = v_b \hat{\boldsymbol{\phi}}.$$

The velocity of the bug in inertial space by [Equation \(15.4\)](#) is

$$\left[\frac{d\mathbf{r}}{dt} \right]_{\text{inertial}} = \left[\frac{d\mathbf{r}}{dt} \right]_{\text{rot}} + \boldsymbol{\Omega}_0 \times \mathbf{r}$$

$$\mathbf{v}_i = \mathbf{v}_{\text{rot}} + \boldsymbol{\Omega}_0 \times \mathbf{r}$$

$$= v_b \hat{\boldsymbol{\phi}} + \Omega_0 r (\hat{\mathbf{k}} \times \hat{\boldsymbol{\rho}})$$

$$= v_b \hat{\boldsymbol{\phi}} + \Omega_0 r \hat{\boldsymbol{\phi}}.$$

For convenience, let us write $v_b \hat{\boldsymbol{\phi}}$ as $\Omega_b r \hat{\boldsymbol{\phi}}$. Then

$$\mathbf{v}_i = (\Omega_b + \Omega_0) r \hat{\boldsymbol{\phi}}.$$

(b) By the definition of centripetal acceleration

$$\begin{aligned} \mathbf{a}_i &= \frac{v_i^2}{r} (-\hat{\boldsymbol{\rho}}) = \frac{[(\Omega_b + \Omega_0)r]^2}{r} (-\hat{\boldsymbol{\rho}}) \\ &= -\hat{\boldsymbol{\rho}} r (\Omega_b + \Omega_0)^2. \end{aligned}$$

(c) Using Equation (15.4) to determine $\mathbf{a}_i$ we have

$$\begin{aligned} \mathbf{a}_i &= \left[\frac{d\mathbf{v}_i}{dt} \right]_{\text{inertial}} = \left[\frac{d}{dt} (\Omega_b + \Omega_0) r \hat{\boldsymbol{\phi}} \right]_{\text{inertial}} \\ &= \left[\frac{d}{dt} (\Omega_b + \Omega_0) r \hat{\boldsymbol{\phi}} \right]_{\text{rot}} + \boldsymbol{\Omega}_0 \times (\Omega_b + \Omega_0) r \hat{\boldsymbol{\phi}}. \end{aligned}$$

Since $v_b = \Omega_b r$ and because the rate of change of $\hat{\boldsymbol{\phi}}$ in the rotating frame is only due to the motion of the bug, we have $\left[\frac{d\hat{\boldsymbol{\phi}}}{dt} \right]_{\text{rot}} = -\Omega_b \hat{\boldsymbol{\rho}}$. Furthermore, keeping in mind that $|\mathbf{r}| = r = \text{constant}$, we obtain

$$\begin{aligned} \mathbf{a}_i &= (\Omega_b + \Omega_0) r (-\Omega_b) \hat{\boldsymbol{\rho}} + \Omega_0 (\Omega_b + \Omega_0) r (\hat{\mathbf{k}} \times \hat{\boldsymbol{\phi}}) \\ &= -\hat{\boldsymbol{\rho}} r [(\Omega_b + \Omega_0) \Omega_b + \Omega_0 (\Omega_b + \Omega_0)] \\ &= -\hat{\boldsymbol{\rho}} r [\Omega_b^2 + 2\Omega_b \Omega_0 + \Omega_0^2] \\ &= -\hat{\boldsymbol{\rho}} r (\Omega_b + \Omega_0)^2, \end{aligned}$$

in agreement with the result of part (b).

15.4 Centrifugal Force and the Plumb Bob

Assume the Earth is a sphere. Most people probably assume that a hanging plumb bob points directly towards the center of the Earth. However, that would only be true if the Earth were not

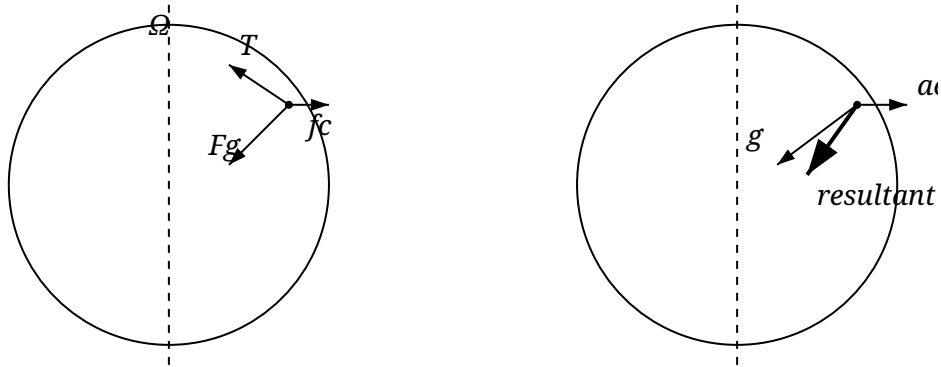


Figure 15.4 A plumb bob in the Northern Hemisphere points below the center of a spherical Earth. The sketch on the left shows the "forces" acting on the bob. The vector $\mathbf{f}_c$ represents the centrifugal force. The sketch on the right illustrates the vector sum of $\mathbf{g}$ and $\mathbf{a}_c = -\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$.

rotating. On a rotating Earth, the centrifugal force causes the plumb bob to be deflected slightly away from the line to the center. The proof is fairly simple. You may want to try it on your own before reading my derivation.

As shown in [Figure 15.4](#), the "real" forces acting on the bob are the tension in the string and the gravitational attraction of the Earth. That is,

$$\mathbf{F} = \mathbf{T} + m\mathbf{g}.$$

Inserting this expression for the force into the " $\mathbf{F} = m\mathbf{a}$ " equation in the form of [Equation \(15.6\)](#) we get

$$\mathbf{T} + m\mathbf{g} - 2m\boldsymbol{\Omega} \times \mathbf{v}_r - m[\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})] = m\mathbf{a}_r,$$

or

$$\mathbf{T} + m\mathbf{g} - m[\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})] = 0,$$

where I used the fact that the velocity and acceleration of the plumb bob in the rotating frame are zero.

Now $\mathbf{g}$ points towards the center of the Earth, but the equation for the plumb bob written in the form

$$\mathbf{T} + m\mathbf{g} = m[\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})],$$

indicates that the tension is not parallel to $m\mathbf{g}$ and therefore it is not directed along the line to the center of the Earth.

It is usually convenient to include the centrifugal acceleration into the gravitational

acceleration. The "effective" gravitational acceleration is defined by

$$\mathbf{g}_e = \mathbf{g} - \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}),$$

and the equation reduces to

$$\mathbf{T} + m\mathbf{g}_e = 0.$$

That is, the tension is directed opposite to $\mathbf{g}_e$ rather than opposite to $\mathbf{g}$. See [Figure 15.4](#). In the Northern Hemisphere $\mathbf{g}_e = \mathbf{g} - \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$ points slightly below the center of the Earth while in the Southern Hemisphere it points slightly above the center.

The centrifugal force is responsible for the fact that the Earth is not a sphere; it has a "bulge" at the equator. To see how this comes about, consider a particle on the surface of a perfectly spherical Earth. This particle is subjected to the gravitational force, a normal force and the centrifugal

force. If you draw a force diagram for this situation, you will find that the centrifugal force is an unbalanced force. Resolving the centrifugal force into components parallel and perpendicular to the Earth's surface, you will find that the perpendicular component simply adds to the normal force, but the unbalanced parallel component points toward the equator. Thus, the centrifugal force is responsible for the fact that the Earth is an oblate spheroid rather than a sphere. You can convince yourself that if the Earth's equatorial bulge is taken into consideration there will be no unbalanced forces on a particle on the surface. This argument also tells you that although a plumb line does not point towards the center of the Earth, it is perpendicular to the Earth's *surface*, so it does define a local perpendicular.

Exercise 15.3

Draw the force diagram for a particle on the surface of a perfectly smooth, spherical, rotating planet and show that the particle accelerates towards the equator. Do this for both the Northern and Southern Hemispheres. Draw the same sort of sketch for a planet with an equatorial bulge and demonstrate that in this case there is no net force on the particle.

15.5 The Coriolis Force

It is well known (at least to physicists) that a projectile fired near the surface of the Earth will veer towards the right in the Northern Hemisphere and towards the left in the Southern Hemisphere. This is due to the so-called Coriolis force, which is also responsible for oceanic circulation, cyclones, hurricanes, and other geophysical phenomena.

The Coriolis force is the term $-2m\boldsymbol{\Omega} \times \mathbf{v}_r$ in Equation (15.6). Note that the direction of this fictitious force depends on the direction of the velocity vector.

When studying the effect of the Coriolis force on objects moving near the surface of the Earth it is convenient to introduce a Cartesian coordinate system in which the z -axis points up from the surface (i.e., opposite to the direction of the effective gravitational acceleration $\mathbf{g}_e$), the x -axis points East, and the y -axis points North.² See Figure 15.5. Note that the angular velocity of the Earth ($\boldsymbol{\Omega}$) has components along the y -axis (North) and along the $+z$ -axis (Up). Thus:

$$\boldsymbol{\Omega} = \Omega \cos \lambda \hat{\mathbf{j}} + \Omega \sin \lambda \hat{\mathbf{j}},$$

where λ is the "colatitude."

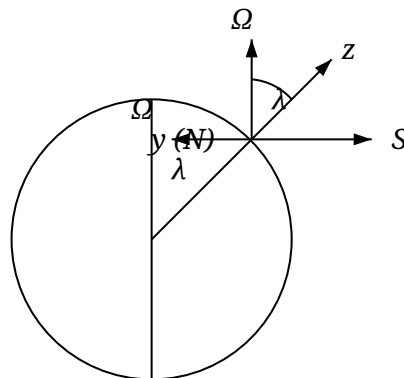


Figure 15.5 A reference frame fixed to the rotating Earth. The z -axis is along the local vertical, the y -axis points North and the x -axis points East (into the page). The angle λ is the colatitude.

² Some books use a different convention with the x -axis pointing South. The equations for the motion will be slightly different, so be careful when comparing equations from different books.

Exercise 15.4

Draw force diagrams to determine the direction of the Coriolis force on the following objects. (a) A falling stone. (b) A projectile fired due North in the Northern Hemisphere. (c) A projectile fired due East in the Northern Hemisphere. (d) A projectile fired due East in the Southern Hemisphere.

15.5.1 A Falling Body

An object of mass m is dropped from a height h . Where does it land? It turns out that the object hits the ground a tiny bit to the East of the spot directly below the point where it was released. The reason is that the falling object has a downward velocity so there is a Coriolis force acting on it, given by $-2m\boldsymbol{\Omega} \times \mathbf{v}_r$. (In the rotating reference frame of the Earth, the forces acting on the particle are the gravitational force, the Coriolis force, and the centrifugal force.)

For simplicity, we incorporate the centrifugal acceleration into $\mathbf{g}$ and write $\mathbf{g}_e$ for the sum of these two terms. The equation of motion of the falling body is, then

$$m\mathbf{a}_r = m\mathbf{g}_e - 2m\boldsymbol{\Omega} \times \mathbf{v}_r.$$

If you think about it, you will soon realize we cannot solve this equation for $\mathbf{a}_r$, because we do not know $\mathbf{v}_r$, and we cannot determine $\mathbf{v}_r$ until we solve the equation! It would seem that we are faced with an unsolvable problem.

The way out of this dilemma is to use the fact that the Coriolis acceleration is much smaller than the gravitational acceleration, so $\mathbf{v}_r$ is almost equal to the velocity of the falling body in the absence of the Coriolis force. That is, the Coriolis effect can be treated as a *perturbation*. The technique is as follows: First solve the unperturbed problem and determine the unperturbed velocity and time for the object to fall. This is called the "zeroth-order approximation." Then use the unperturbed velocity in the expression for the Coriolis acceleration to obtain the "first-order approximation." This process can be repeated as many times as desired to get better and better approximations to its velocity. (A first- or second-order approximation is quite adequate.)³

Let us begin by solving the problem for the *unperturbed* velocity and position of the particle as a function of time (the zeroth-order approximation). The only force acting on the body is the (effective) gravitational force. Consequently,

$$\mathbf{a}_r = -g_e \hat{\mathbf{k}}.$$

Integrating once,

$$\mathbf{v}_r = -g_e t \hat{\mathbf{k}},$$

and integrating again,

$$\mathbf{z} = h \hat{\mathbf{k}} - \frac{1}{2} g_e t^2 \hat{\mathbf{k}}.$$

The time required for the particle to reach the ground, t_f , is the value of t when $z = 0$, that is,

$$t_f = \sqrt{2h/g_e}.$$

Note that none of the above relations include the Coriolis effect.

Now include the Coriolis acceleration, given by:

$$\mathbf{a}_c = -2\boldsymbol{\Omega} \times \mathbf{v}_r.$$

³ This technique is known as the method of successive approximations.

You cannot evaluate $\mathbf{a}_c$ exactly because you do not know the value of $\mathbf{v}_r$ which itself depends on the Coriolis acceleration. However, you know the zeroth-order approximation to the velocity. It is $\mathbf{v}_r = -g_e t \hat{\mathbf{k}}$. The *first-order* correction is obtained by including the Coriolis term in the acceleration but using the zeroth-order velocity. That is, $\mathbf{a} = -g_e \hat{\mathbf{k}} + \mathbf{a}_c$ where

$$\begin{aligned} \mathbf{a}_c &= -2\boldsymbol{\Omega} \times \mathbf{v}_r = -2[\Omega \cos \lambda \hat{\mathbf{k}} + \Omega \sin \lambda \hat{\mathbf{j}}] \times [-g_e t \hat{\mathbf{k}}] \\ &= 2g_e t \Omega \sin \lambda \hat{\mathbf{i}}. \end{aligned}$$

Note that for the falling body, the Coriolis acceleration is in the positive x -direction, that is, towards the East. The eastward component of the velocity (v_x) is initially zero and at any later time it is obtained by integrating the expression above. Therefore,

$$v_x = g_e t^2 \Omega \sin \lambda.$$

The eastward deflection (x) is obtained by integrating again.

$$x = \frac{1}{3}g_e t^3 \Omega \sin \lambda.$$

To determine the total eastward deflection when the object is dropped from height h , substitute the time of flight ($t = \sqrt{2h/g_e}$) into this last expression.

The results obtained are the first-order solution. As mentioned above, these results can be plugged back into the equation of motion to obtain the second-order solution, and so on.

Students are sometimes mystified by this result. The eastward deflection is not due to the Earth rotating under the falling body – anyway, that would give a westward deflection! A physical explanation for the horizontal (eastward) deflection of a dropped body comes from the conservation of angular momentum. Let $\mathbf{l}$ be the angular momentum of the particle in the inertial frame. There is no torque acting on it, so $\mathbf{l}$ is constant. Initially the object is at position $\mathbf{r}$ (where $\mathbf{r}$ is a vector from the center of the Earth). In the inertial frame the body has an eastward linear velocity $\mathbf{v}$. Because the body is at rest with respect to the surface of the Earth, this initial velocity is $\mathbf{v} = \boldsymbol{\Omega} \times \mathbf{r} \doteq \Omega R_e \sin \lambda \hat{\mathbf{i}}$. The angular momentum of the particle is

$$\mathbf{l} = m\mathbf{r} \times \mathbf{v}.$$

For $\mathbf{l}$ to remain constant while $\mathbf{r}$ decreases (as the particle falls), the velocity $\mathbf{v}$ must increase. Therefore, in the Earth-based reference frame, the object acquires a velocity towards the East.⁴

Another strange and interesting effect of Coriolis forces is the fact that if you throw an object straight up, it does not land at the point from which it started. (You will prove this in [Problem 15.6](#).)

Worked Example 15.2 To first order, a falling particle is not deflected towards the South. However, there is a southward deflection in the second-order approximation. Estimate this deviation and compare it with the first-order deviation towards the East.

Solution To first order, the velocity is

$$\mathbf{v}_r = v_x \hat{\mathbf{i}} + v_z \hat{\mathbf{k}} = -g_e t \hat{\mathbf{k}} + g_e t^2 \Omega \sin \lambda \hat{\mathbf{i}}.$$

⁴ A slightly different explanation for the eastward deflection of a dropped object is based on the fact that if you hold an object above a point on the Earth's surface, both the object and the surface point are moving towards the East with the same angular velocity (Ω). But because $v = \Omega r$ the object has a slightly greater eastward velocity than the surface point. This greater eastward velocity is not affected by the fall and therefore the object lands to the East of the surface point.

The Coriolis acceleration is

$$\begin{aligned}\mathbf{a}_c &= -2\boldsymbol{\Omega} \times \mathbf{v}_r \\ &= -2 \left[\Omega \cos \lambda \hat{\mathbf{k}} + \Omega \sin \lambda \hat{\mathbf{j}} \right] \times \left[-g_e t \hat{\mathbf{k}} + g_e t^2 \Omega \sin \lambda \hat{\mathbf{i}} \right].\end{aligned}$$

That is,

$$\begin{aligned}a_x &= 2(\Omega \sin \lambda) g_e t, \\ a_y &= -2(\Omega^2 \cos \lambda \sin \lambda) g_e t^2, \\ a_z &= 2(\Omega^2 \sin^2 \lambda) g_e t^2 - g_e.\end{aligned}$$

Consequently,

$$v_y = \int a_y dt = -\frac{2}{3}(\Omega^2 \cos \lambda \sin \lambda) g_e t^3,$$

and

$$y = -\frac{1}{6}(\Omega^2 \cos \lambda \sin \lambda) g_e t^4.$$

Using the first-order value for the time of flight we obtain

$$y = -\frac{2h^2}{3g_e} \Omega^2 \cos \lambda \sin \lambda.$$

Note that the time of flight will actually be slightly longer because the Coriolis acceleration opposes the gravitational acceleration.

Comparing the southward and eastward deflections we have

$$\frac{y}{x} = \frac{-(1/6)(\Omega^2 \cos \lambda \sin \lambda) g_e t^4}{(1/3)(\Omega \sin \lambda) g_e t^3} = -\frac{1}{2}(\Omega \cos \lambda) t = -\Omega \cos \lambda \sqrt{h/2g_e}.$$

Since Ω is about 10^{-5} and $\cos \lambda$ is near unity and the time of flight will be in the range of 1–10 s, we appreciate that the southward deflection is significantly smaller than the eastward deflection.

Exercise 15.5

A stone is dropped from a height of 50 m. How far is it deflected towards the

East? Assume the latitude is 60° N. Answer: 0.39 cm.

15.5.2 A Projectile

To determine the motion of a projectile fired from the surface of the Earth with arbitrary initial speed and direction, let us return to the equation for the acceleration in a rotating coordinate frame. A slight modification of [Equation \(15.5\)](#) gives

$$\mathbf{a} = \mathbf{g} - 2(\boldsymbol{\Omega} \times \mathbf{v}).$$

I dropped the subscript "e" on $\mathbf{g}$, but keep in mind that the centrifugal force is incorporated into $\mathbf{g}$. I also dropped the subscript r on $\mathbf{v}_r$ and on $\mathbf{a}_r$. The velocity $\mathbf{v}$ at any time can be expressed as

$$\mathbf{v} = v_x \hat{\mathbf{i}} + v_y \hat{\mathbf{j}} + v_z \hat{\mathbf{k}},$$

and

$$\boldsymbol{\Omega} = \Omega \sin \lambda \hat{\mathbf{j}} + \Omega \cos \lambda \hat{\mathbf{k}}.$$

Hence,

$$\mathbf{a} = -g \hat{\mathbf{k}} + 2(v_y \Omega \cos \lambda - v_z \Omega \sin \lambda) \hat{\mathbf{i}} - 2v_x \Omega \cos \lambda \hat{\mathbf{j}} + 2v_x \Omega \sin \lambda \hat{\mathbf{k}}, \quad (15.7)$$

or

$$a_x = 2v_y \Omega \cos \lambda - 2v_z \Omega \sin \lambda,$$

$$a_y = -2v_x \Omega \cos \lambda,$$

$$a_z = 2v_x \Omega \sin \lambda - g.$$

Recall that you cannot determine the motion by integrating the acceleration terms twice because the accelerations depend on the velocity which is unknown. So once again it is necessary to use the method of successive approximations. That is, you obtain the first-order approximation by using the velocity computed by using the zeroth-order approximation, and similarly calculating the higher-order approximations. However, even without doing any calculations, the functional

dependence of the acceleration components gives a good qualitative idea of the motion. For example, in the Northern Hemisphere, if the projectile is fired due North, then v_x (the eastward component of the velocity) is initially zero and there is no Coriolis acceleration in the y - or z -directions. There is, however, an eastward component of the acceleration and the projectile will be deflected towards the East (to the right of its direction of motion). To actually evaluate this deflection you must incorporate the zeroth-order approximations of v_y and v_z into the expression for a_x and then integrate.

Similarly, the expressions for the acceleration components indicate that a projectile fired due East (x -direction) is deflected towards the South. The velocity component v_x slightly changes a_z , making it less negative, therefore slightly increasing the time of flight of the projectile resulting in an increase in range. Furthermore, a_x is nonzero, which also affects the range of the projectile. Although the vertical velocity v_z is positive during the first part of the trajectory and negative during the second part, you cannot assume that the second term in a_x averages out to zero.

In solving for the motion of a projectile, you must use perturbation techniques, but fortunately a reasonably accurate solution is usually given by the lowest order approximation that includes the Coriolis force.

You should convince yourself that in the Northern Hemisphere a projectile will be deflected toward its right and in the Southern Hemisphere it will be deflected toward its left.

Worked Example 15.3 A large cannon is located at colatitude λ in the Northern Hemisphere. It fires a shell with velocity v_0 . The cannon is aimed at an angle ϕ East of North and has an elevation angle θ . (a) Write expressions for the initial (unperturbed) velocity components. (Call them v_{0x}, v_{0y}, v_{0z} .) (b) Write expressions for a_x, a_y, a_z in terms of the unperturbed velocity components. (c) Obtain first-order expressions for the velocity components as functions of time and the unperturbed values of the velocity. Ignore air resistance. (The results of this example will be found useful when solving the problems at the end of the chapter.)

Solution (a) By inspection of [Figure 15.6](#) the initial velocity components are

$$v_{0x} = v_0 \cos \theta \sin \phi,$$

$$v_{0y} = v_0 \cos \theta \cos \phi,$$

$$v_{0z} = v_0 \sin \theta.$$

